Additive Manufacturing
Cost Model

Methodology, Formula Reference & Case Study

7	6+	5	500
Cost Categories	Process Steps	Extension Modules	Quantity Curve Points

This document describes the complete cost estimation methodology used by the AM Cost Calculator. It covers how each cost driver is modelled — from machine depreciation and powder consumption to support removal labour — and walks through a fully worked case study for a 316L stainless steel medical impeller produced in a batch of 100 parts via Laser Powder Bed Fusion (LPBF). Intended for program managers, procurement teams, and business stakeholders who need to understand and validate cost estimates without a background in additive manufacturing.

Version	Date	Classification	Process
1.0	**April 2026**	**Business Use**	**LPBF Metal AM**

1 What This Tool Does

This calculator estimates the total and per-part cost of producing metal parts using **Laser Powder Bed Fusion (LPBF)** — the most widely used industrial metal additive manufacturing process. The methodology follows the framework developed at MIT and used by leading AM cost researchers, adapted for real-world production planning.

Input Type	What It Captures
Part Geometry	Size, volume, height, support structures — from a 3D file or manual entry
Machine & Facility	Which printer, its capital cost, operating schedule, staffing model
Material	Powder type, cost per kilogram, density, layer thickness

The tool computes cost across **seven distinct categories**, each described in detail below.

2 The LPBF Process — Step by Step

Phase	Description
1. Powder spreading	A thin metal powder layer (30–60 µm, ~width of a human hair) is spread across the build plate by a recoater blade.
2. Laser melting	One or more laser beams selectively melt the powder, tracing each part's cross-section for that layer.
3. Layer repeat	The build plate drops by one layer thickness. Steps 1–2 repeat thousands of times.
4. Cool-down	After the final layer, the machine undergoes a controlled cooling phase to prevent distortion.
5. Part removal	The build plate is removed and parts are separated (wire EDM or sawing).
6. Post-processing	Support structures are manually removed. Optional: heat treatment, HIP, surface finishing, NDT inspection.

Every one of these steps carries a cost. The model accounts for all of them — including machine time during cool-down, operator time during changeover, and technician time removing supports after the build.

3 Cost Categories & Formulas

The total cost is the sum of seven categories. Each is calculated independently and can be inspected in the tool's results breakdown.

#	Category	Primary Driver	Typical Share
1	Material	Part volume, powder price, recycling rate	10–20%
2	Build Preparation	Number of builds, engineer salary	3–6%
3	Machine Usage	Machine hourly rate × total machine hours	20–35%
4	Consumables	Inert gas duration, per-build supplies	4–8%
5	Labor	Operator supervision + build changeover time	8–15%
6	Support Removal	Part geometry complexity, technician rate	15–30%
7	Extensions	Heat treatment, part separation, HIP, NDT	0–25%

3.1 Material

The recoater spreads powder over the *entire* build plate for every layer — not just where parts sit. Most unfused powder is recycled, but a fraction degrades and becomes waste.

```
Powder deposited = Num builds × Build plate area × Part height × Powder density
Unfused powder = Powder deposited − Part material − Support material
Material waste = Unfused powder × (1 − Recycling fraction)
Material purchased= Part material + Support material + Material waste
Material cost = Material purchased × Cost per kg
```

A large build plate with few small parts wastes a disproportionate amount of powder. Filling the build plate efficiently is the single most effective way to reduce material cost.

3.2 Build Preparation

An engineer sets up each build: orienting parts, generating supports, slicing the file, and configuring process parameters. The first build takes longer due to full process verification.

```
Prep time = First build time + (Num builds − 1) × Subsequent build time
Cost = Prep time × Engineer hourly rate
```

3.3 Machine Usage

Machine cost is converted from a capital purchase into an hourly rate using the **equivalent annual cost (EAC)** method — standard engineering economics practice. The hourly rate is then multiplied by total machine-on time across all builds.

```
Annual depreciation = (CoC × Purchase price × Infrastructure factor)
÷ (1 − (1 + CoC)^(−Lifetime years))
Annual maintenance = Purchase price × Maintenance %
Machine $/hr = (Depreciation + Maintenance)
÷ (Hrs/day × Days/wk × 52 × Uptime %)
Machine cost = Total machine time × Machine $/hr
```

Time Phase	Formula	Notes
Warm-up	Num builds × Warm-up hr/build	Machine heats before each build
Laser scanning	Production run × Volume × (1+Support%) ÷ (Rate × Lasers)	Scales with material volume & laser count
Recoating	Scan time × (1/Recoating efficiency – 1)	Overhead between laser passes
Cool-down	Num builds × Cooldown hr/build	Controlled cooling to prevent distortion
Build exchange	Num builds × (Setup + Removal) hr/build	Plate swap between builds

3.4 Consumables

Inert gas (argon or nitrogen) continuously purges the build chamber to prevent metal oxidation. Per-build supplies include filters, recoater blades, and build platform material.

```
Inert gas cost = Gas $/hr × (Warm-up + Print + Cool-down time)
Per-build cost = Consumable $/build × Num builds
Total consumables = Inert gas + Per-build cost
```

3.5 Labor

Labor covers two distinct activities: monitoring the machine while it runs, and physically exchanging build plates between builds. The FTE supervision fraction reflects that one operator can monitor multiple machines simultaneously.

```
Labor during build = (Warm-up + Print + Cool-down) × FTE supervised
Labor at changeover = Build exchange time × FTE for changeover
Labor cost = Total labor hours × Operator hourly rate
```

3.6 Support Removal (Post-Processing)

Support structures are temporary scaffolding printed alongside the part to anchor overhangs. They must be manually removed after printing. The time scales with the square root of the part's projected footprint area — larger parts have more interface to remove, but not linearly.

```
Removal time/part = Support constant × √(Part projected area)
Total removal time = Time per part × Num parts
Post-process cost = Total removal time × Technician hourly rate
```

Support removal is frequently the largest single cost driver for geometrically complex parts. Redesigning to minimise overhangs, or rotating the build orientation, can cut this cost by 30–60%.

3.7 Metal AM Extensions (Optional)

These post-build processes are common in metal AM production and can be toggled on or off. They are costed separately so their contribution is fully transparent.

Process	Why It's Done	How Costed
Heat Treatment	Relieve residual stress; restore ductility	Fixed $ per build
Part Separation	Cut parts off build plate (wire EDM / saw)	$ per part
HIP	Eliminate internal porosity for safety-critical use	$ per part
Surface Treatment	Machining, shot peening, electropolishing	$ per part
NDT Inspection	CT scan, X-ray, or ultrasonic testing	$ per part

4 Case Study — Medical Stainless Steel Impeller

A fluid-handling impeller for a medical device, manufactured in **316L Stainless Steel** via LPBF. The part contains internal flow channels that cannot be produced by conventional machining, making additive manufacturing the only viable production route.

Part Parameter	Value	Machine Parameter	Value
Material	316L Stainless Steel	Machine cost	$600,000
Part mass	180 g	Lifetime	10 years
Part volume	22.5 cm^3	Cost of capital	7%
Build height	90 mm	Infrastructure	12% of purchase
Build footprint	30 cm^2	Maintenance	7% of purchase
Support fraction	18% of part mass	Build plate area	784 cm^2 (280×280mm)
Production run	100 parts	Build rate	15 cm^3/hr (1 laser)
		Machine uptime	80%

Material Parameter	Value
Powder cost	$40 / kg
Density	8.0 g/cm^3
Packing policy	2D (single layer)
Packing efficiency	55%
Powder recycling	80%
Post-processing	Heat treatment ($300/build) · Part separation ($50/part)

4.1 Build Planning

```
Parts per build = floor( 784 cm² ÷ 30 cm² × 0.55 ) = floor(14.37) = 14 parts
Number of builds = ceil( 100 ÷ 14 ) = 8 builds
```

14 parts fit per build · 100 parts require 8 builds

4.2 Material Cost

```
Part material (100 parts) : 100 × 0.180 kg = 18.00 kg
Support material (18%) : 100 × 0.032 kg = 3.24 kg
Powder deposited (8 builds): 8 × 784 cm² × 9.0 cm × 0.008 = 451.60 kg
Unfused powder : 451.60 – 18.00 – 3.24 = 430.36 kg
Recycled (80%) : = 344.29 kg
Wasted (20% of unfused) : = 86.07 kg
Material purchased : 18.00 + 3.24 + 86.07 = 107.31 kg
Material cost : 107.31 kg × $40/kg = $4,292
```

451 kg of powder enters the machine across 8 builds, but 80% of unfused powder is recycled. Only 107 kg needs to be purchased — a 76% reduction vs gross consumption.

Material cost: $4,292 ($42.92/part)

4.3 Build Preparation

```
Prep time = 5 hr (first build) + 7 × 2 hr (subsequent) = 19 hr
Cost = 19 hr × $75/hr (engineer) = $1,425
```

Build prep cost: $1,425 ($14.25/part)

4.4 Machine Usage

```
Total upfront cost : $600,000 × 1.12 = $672,000
Annual depreciation : 0.07 × $672,000 ÷ (1 – 1.07^–10) = $95,671/yr
Annual maintenance : $600,000 × 7% = $42,000/yr
Annual machine cost : $95,671 + $42,000 = $137,671/yr
Operating hours/yr : 20 hr/day × 5 days × 52 wks × 80% = 4,160 hr/yr
Machine cost/hr : $137,671 ÷ 4,160 = $33.09/hr
```

Phase	Calculation	Time
Warm-up	8 builds × 2 hr	16 hr
Laser scanning	100 × 22.5 cm³ × 1.18 ÷ 15 cm³/hr	177 hr
Recoating	177 hr × (1/0.75 – 1)	59 hr
Cool-down	8 builds × 4 hr	32 hr
Build exchange	8 builds × 1.5 hr	12 hr
Total		**296 hr**

Machine cost: 296 hr × $33.09/hr = $9,795 ($97.95/part)

4.5 Consumables

```
Build active time (warm-up + print + cool-down) : 16 + 236 + 32 = 284 hr
Inert gas cost : 284 hr × $2/hr = $568
Per-build supplies: 8 builds × $200/build = $1,600
Total consumables : = $2,168
```

Consumables cost: $2,168 ($21.68/part)

4.6 Labor

```
Monitoring (0.25 FTE): 284 hr × 0.25 = 71 hr
Build changeover : 12 hr × 1.0 = 12 hr
Total labor : = 83 hr
Labor cost : 83 hr × $50/hr = $4,150
```

Labor cost: $4,150 ($41.50/part)

4.7 Support Removal

```
Time per part : 0.4 (constant) × √30 cm² = 0.4 × 5.48 = 2.19 hr/part
Total time : 2.19 hr × 100 parts = 219 hr
Cost : 219 hr × $40/hr (technician) = $8,760
```

Post-process cost: $8,760 ($87.60/part)

4.8 Metal AM Extensions

```
Heat treatment : $300/build × 8 builds = $2,400
Part separation : $50/part × 100 parts = $5,000
Extensions total: = $7,400
```

Extensions cost: $7,400 ($74.00/part)

5 Cost Summary & Results

Cost Category	Total Cost	Per Part	Share
1. Material	$4,292	$42.92	11.3%
2. Build Preparation	$1,425	$14.25	3.8%
3. Machine Usage	$9,795	$97.95	25.8%
4. Consumables	$2,168	$21.68	5.7%
5. Labor	$4,150	$41.50	10.9%
6. Support Removal	$8,760	$87.60	23.1%
7. Heat Treatment	$2,400	$24.00	6.3%
8. Part Separation	$5,000	$50.00	13.2%
TOTAL	**$37,990**	**$379.90**	**100.0%**

$379.90	$37,990	14	8	$33.09/hr	107 kg
Cost per Part	Total Run Cost	Parts per Build	Total Builds	Machine Rate	Powder Purchased

6 Key Insights for Decision-Makers

7.60/part · 23%)

This is the most frequent surprise for teams new to metal AM. Reducing
or rotating the build direction, can cut this cost by 30–60%.

share (26%)

The machine costs $33/hr. Running it for 296 hours across 8 builds pr
instead of 14 would reduce machine cost per part by ~30%.

not the main cost

Unlike CNC machining (where buy-to-fly ratio dominates), in AM the mat
a lower-cost alloy saves far less than filling the build plate better.

bined

Heat treatment and part separation are often quoted separately by AM
explicit, attributable, and comparable across machine options.

At 100 parts: $380/part. At 50 parts: ~$490/part. At 200 parts: ~$310/
treatment) amortise over more units at scale.

7 Cost Reduction Levers

Lever	Why It Works	Est. Saving Poter
Design for minimal supports	Cuts support removal labour — often the #1 cost	15–40% reduction

Fill the build plate	Fewer builds → lower machine time, prep, and gas per part	10–30% reduction
Increase production volume	Fixed costs amortise over more parts	10–25% reduction
Choose machines with larger plates	More parts per build without increasing print time	5–20% reduction
Reduce build height	Fewer powder layers → faster print + less recoating time	5–15% reduction
Simplify post-processing	Each extension module adds 5–15% when enabled	Varies per step
Negotiate powder recycling terms	Higher recycling fraction directly reduces material required	3–10% reduction

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This document was generated from the AM Cost Calculator. All calculations follow the methodology described above and can be reproduced by entering the same parameters into the tool. Figures assume standard LPBF process conditions; actual costs may vary based on supplier, location, and specific machine configuration.